

PHASE 3

Project Design Phase

Solution architecture, data flow, and UI/UX design for the PulseView web experience

PulseView – Heart Disease Risk Analysis Dashboard

Project	PulseView – Heart Disease Risk Analysis Dashboard
Document	Project Design Report
Version	1.0
Date	July 2026
Prepared By	Project Team

1. Introduction

This document covers how the PulseView concept from the Ideation and Requirement Analysis phases was translated into a concrete technical design: the overall solution architecture, how data moves from source to screen, and the page-by-page UI design.

2. Solution Architecture

PulseView deliberately keeps the Flask application thin. All data preparation and visualization logic lives in Tableau; Flask's job is to provide the surrounding site (navigation, copy, layout) and embed the two published Tableau views by URL.

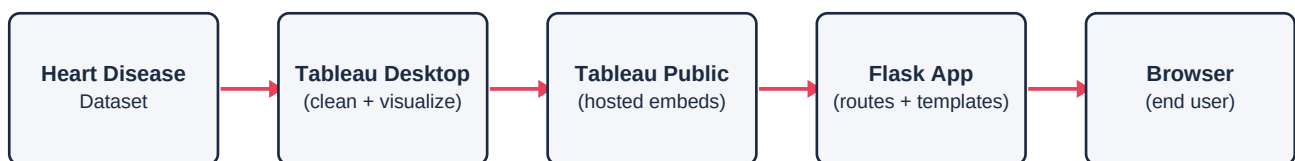


Figure 1: High-level solution architecture

2.1 Application Layer Breakdown

Layer	Technology	Responsibility
Data & visualization	Tableau Desktop / Tableau Public	Cleans the dataset, builds the dashboard and 4-scene story, and hosts them as public, embeddable views.
Application	Flask (Python)	Defines routes (<code>/</code> , <code>/about</code> , <code>/dashboard</code> , <code>/story</code> , <code>/contact</code>), passes structured data (<code>STORY_SCENES</code> , <code>DASHBOARD_PANELS</code>) to templates.
Presentation	Jinja2 templates + CSS + vanilla JS	Renders page layout, story tab navigation, and responsive styling.

3. Data Flow

The two embed URLs (dashboard and story) are defined once, as constants, at the top of `app.py`. Nothing else in the templates needs to change when the underlying Tableau workbook is republished — only those two constants are updated.

- **Step 1:** The raw dataset is loaded and cleaned in Tableau Desktop.
- **Step 2:** The dashboard (4 panels) and the story (4 scenes) are authored as Tableau views and published to Tableau Public.
- **Step 3:** The Flask route handler for `/dashboard` or `/story` renders the matching template, passing the embed URL and the panel/scene copy as context.
- **Step 4:** The template drops the embed URL into an `iframe`; Tableau Public serves the interactive visualization directly to the visitor's browser.

4. UI / UX Design

Five pages make up the site. Each is designed around a single, clear purpose so a first-time visitor is never more than one click from either the story or the dashboard.

4.1 Home

A hero section introduces PulseView and routes visitors toward the story and dashboard.

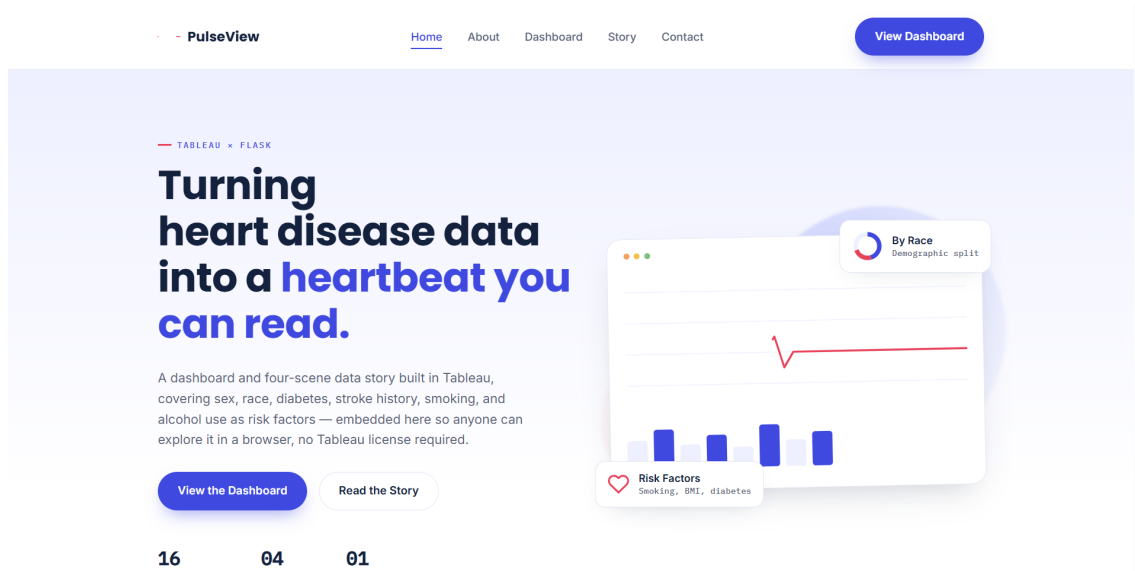


Figure 2: Home page

4.2 About

Explains why the dataset matters, introduces the three personas, and summarises the six-step build process.

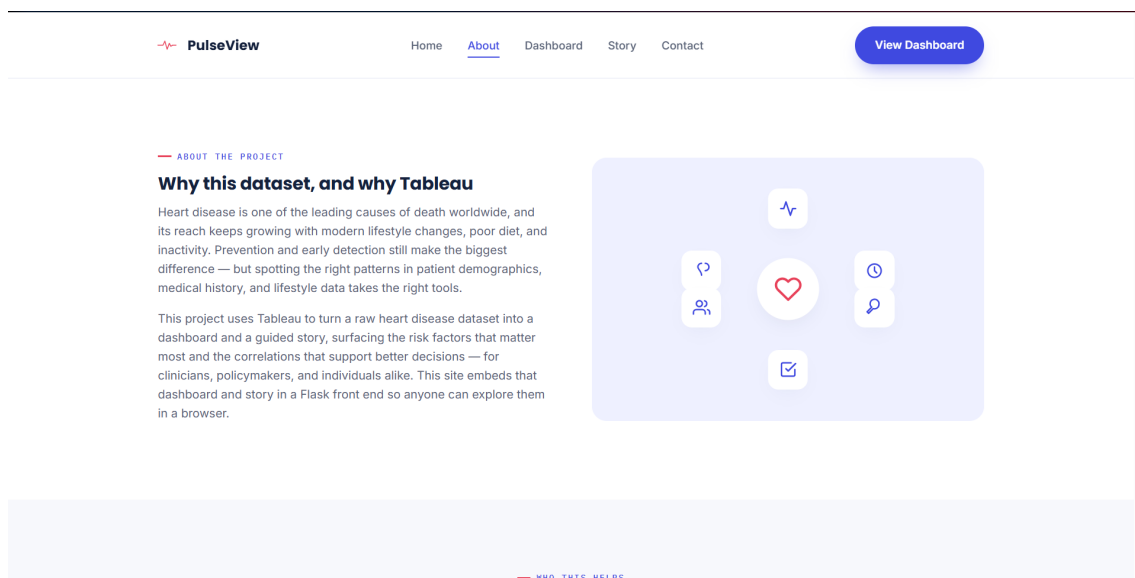


Figure 3: About page

4.3 Dashboard

Embeds the full Tableau dashboard alongside written context for each of the four panels.

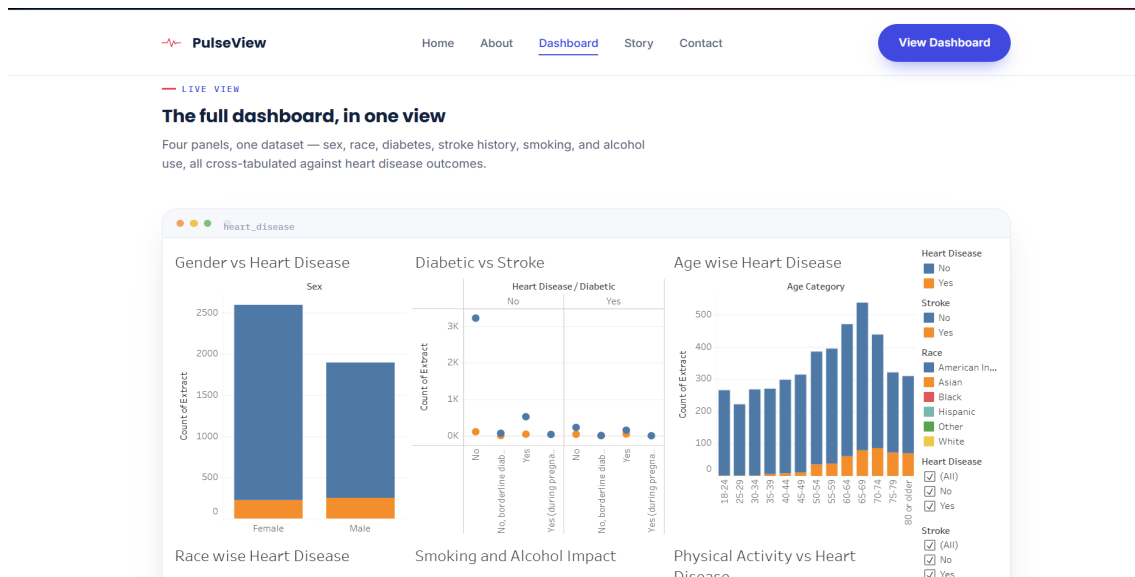


Figure 4: Dashboard page

4.4 Story

A tabbed, four-scene narrative with previous/next controls and keyboard arrow-key support, so the data can be read in a guided order.

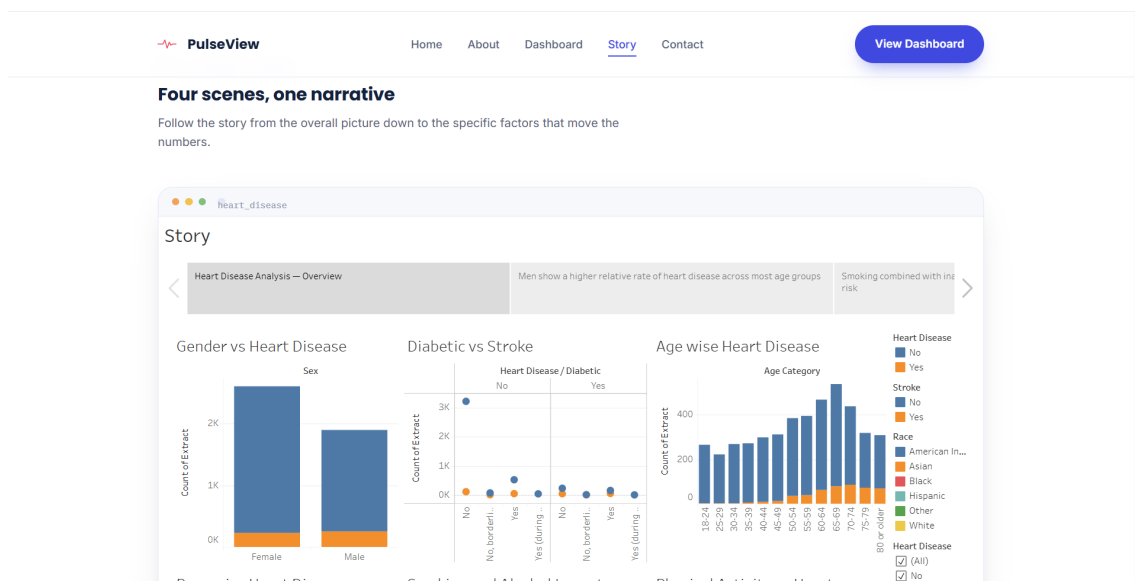


Figure 5: Story page

4.5 Contact

A simple form (name, email, message) with inline validation and flash-message feedback on submit.

PulseView Home About Dashboard Story Contact View Dashboard

— GET IN TOUCH

Questions about the data or the dashboard?

Send a note and we'll get back to you — whether it's about the methodology, the dataset, or extending the dashboard.

Email
hello@pulseview-analytics.example

Project scope
Tableau dashboard & story, embedded in a Flask front end.

Response time
Usually within 2-3 business days.

Name
Your name

Email
you@example.com

Message
What would you like to know?

Send message

Figure 6: Contact page

5. Design Notes

- No database is used — all page copy is defined as Python data structures in `app.py`, keeping the design intentionally lightweight.
- A single base template (`base.html`) supplies shared navigation and footer markup, which every other template extends.
- Colour and iconography (heart motif, orbiting icon ring on the About page) reinforce the health-data theme without relying on stock imagery.